Unity Notes

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Physics

Rigidbody (component)

- Full physics simulation — gravity, forces, momentum, collisions
- Best for: realistic movement, rolling balls, pucks, ragdolls

Rigidbody Settings Cheat Sheet

Setting	What it does
Mass	Weight, affects force reactions
Linear Damping	Slows down movement over time
Angular Damping	Slows down spinning over time
Is Kinematic	Ignores physics forces, moved only by code/transform
Collision Detection	Discrete (default) vs Continuous/Continuous Dynamic (fast objects)
Freeze Position/Rotation	Locks specific axes

Character Controller (component)

- No physics, manual movement control
- Best for: precise player movement (FPS, platformers)

Input Management

Input Manager (absolute banger)

- The new Input System package
- Uses .inputactions asset, Action Maps, event-driven callbacks (OnMove, OnJump, etc.)
- Supports rebinding, multiple devices, better overall

Input (old one)

- UnityEngine.Input.GetKey(), Input.GetAxis()
- Polling-based, simple, but no built-in rebinding or multi-device support

Camera Follow

Code

- Bad — manual transform.position / Quaternion math to follow the player
- Full control but painful to get right (tunneling, distortion, jitter — all the issues we debugged this whole conversation)

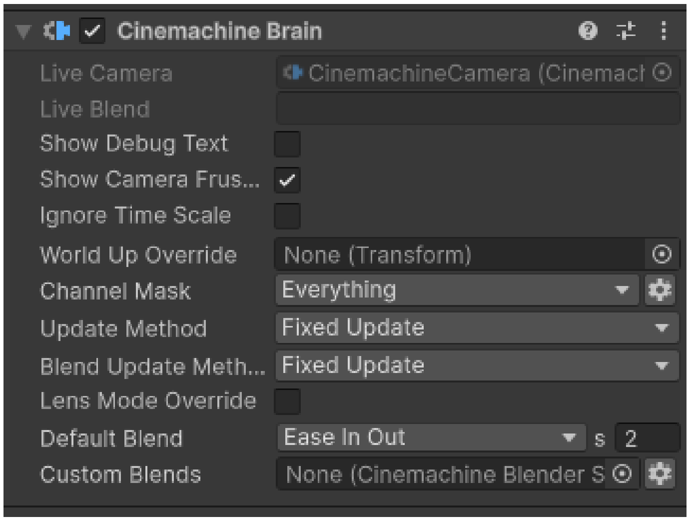
Camera as child object on Player

- Simplest method — just parent the camera in Hierarchy
- Automatically follows position and rotation
- Downside: less flexible for camera switching/transitions

Cinemachine (The best imo) (installed through Package Manager)

- Unity's dedicated camera system package
- Handles follow, look-at, collision avoidance, smooth transitions, shake, and more — all without custom code
- Industry standard for camera work in Unity, used in most professional projects

Note: Be sure to set **Update Method** and **Blend Update Method** to "**Fixed Update**" on the **Cinemachine Brain** component of the **main camera** after adding a CinemachineCamera. It makes the movement much smoother.



Prefabs are useful. *"Prefab everything that you can prefab"*

Prefab in other words is saving an object. It is better for **duplicating**, **better managing** objects, applying **group changes** to objects, etc.

Empty Parent Object trick

Put the model you want to attach code to as a child of an empty object. Changing the model after some time won't cause any loss since all the data are on the empty parent object.

NOTE: be sure to put the **parent object's position** at the **center** (**center of mass**) of the **child object**. That will fix any rotational bug.

Unity Order of Execution

- Awake() → runs once, when object is first created
- OnEnable() → runs every time the object becomes active
- Start() → runs once, after Awake, before first Update
- Update() → runs every frame

FixedUpdate() vs Update()

- **Update()** is recommended to be used for **game logic**. Must use **Time.DeltaTime** for this.
- **FixedUpdate()** is recommended to be used for **Physics**. Must use **Time.FixedDeltaTime** for this.

AddForce() vs AddRelativeForce()

- **AddForce()** adds a **global** force (relateive to the **world's x, y or z** axis)
- **AddRelativeForce()** adds a **Local** force (relative to **the object's x, y and z** axes)

Useful Scripts

- **FindAnyObjectByType<T>()** : Finds an object automatically without dragging into Inspector
- **GetComponent<T>()** : Finding and getting objects/components (on the same object)
- **GetComponentInChildren<T>()**
- **GetComponentInParent<T>()**
- **Destroy(gameObject)**
- **Destroy(gameObject, 3f)** : destroy after delay
- **gameObject.SetActive(false)** : disables/enables the object
- **gameObject.CompareTag("Enemy")**
- **gameObject.layer = LayerMask.NameToLayer("Ground")**

Caching (concept)

The **variables** we have on the top outside of the methods (in each script) is considered as "cache".

Attributes ([something])

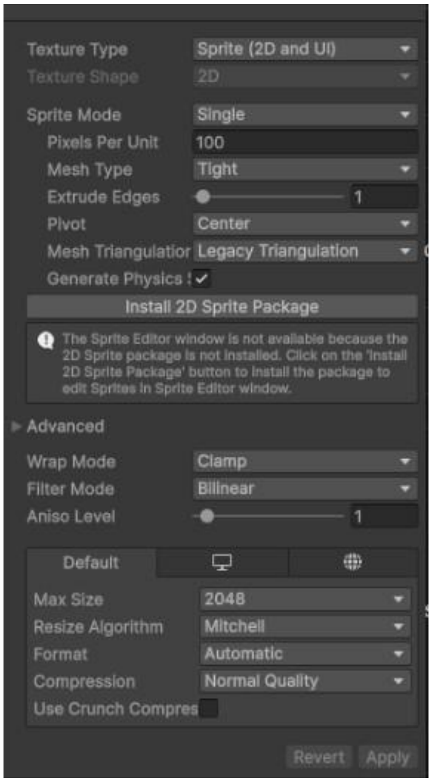
- **[SerializeField]** → Shows a private variable in the Inspector.
- **[HideInInspector]** → Hides a public variable from the Inspector.
- **[Header("Text")]** → Adds a bold section title above the next variable(s) in the Inspector.
- **[Space]** → Adds blank vertical space in the Inspector between variables
- **[Tooltip("text")]** → A tooltip explaining what the variable does when you hover over it in the Inspector
- **[Range(min, max)]** → Turns a number field into a slider between min and max.
- **[Min(value)]** → Prevents a number from going below the specified minimum (no upper limit, no slider)

UI (Canvas)

A canvas must be the parent holding UI elements.

Canvas Scalar better be: "**Scale With Screen Size**" + "**Match Width Or Height**"

- For text: **TextMeshPro** is much better with many more options.
- For images: **Panel** (set sprite image onto it) ("**Set Native Size**" is really useful)



tranform.Rotate()

transform in general can be used to manipulate the object's **position**, **rotation**, and **scale**.

transform.Rotate() is used for rotating in any of the axes. (takes a Vector3 or (x, y, z) as input)

rb.freezeRotation = true/false; (rigidbody)

- **false:** completely disables the object's rotation momentum.
- **true:** enables the object to rotate and have momentum.

Audio Listener (component)

Is a component on the **Main Camera** object. We can setup a **surround sound** too since the main camera has the "ears" to hear sounds and Unity understands where sounds come from automatically.

Audio Source (component)

Has to be added where the audio will be played. Has options like "**Play On Awake**" and "**Loop**".

SceneManager (namespace)

Is used for switching between scenes. Scenes must be added to the project **Scene List** through:

File → Build Profiles → Scene List → Can add the current scene or drag them from project folder.

- **SceneManager.GetActiveScene().buildIndex** → current scene's index (in project's Scene List)
- **SceneManager.sceneCountInBuildSettings** → total number of scenes
- **SceneManager.LoadScene(int sceneIndex)** → to load scene using index (int) or scene name (string)

RAYCAST (IS THE MOST AMAZING THING I'VE EVER SEEN)

It shoots an **invisible line** from a point in a direction and tells you what it hits first.
Used for:

- Player "look at object to interact" (E to pick up, open doors, etc.)
- Shooting/hit detection
- Ground cheks
- AI sight
- Etc.

How to use it for object interaction:
You must set the **Layer** to "**Interactable**" on the target objects. And other Layers for other use cases. For instance, for AI bots to see player, the raycast must check if the collision has occurred to an item with the **Layer** being "**Player**"

```
void Start()
{
    // Get the Camera component from the PlayerLook script (so raycast starts from where player is looking)
    cam = GetComponent<PlayerLook>().cam;

    // Get reference to the UI script that displays interaction prompts
    playerUI = GetComponent<PlayerUI>();

    // Get reference to the Input Manager script that reads player input
    inputManager = GetComponent<InputManager>();
}

void Update()
{
    // Clear the prompt text every frame first (resets if nothing is being looked at)
    playerUI.UpdateText(string.Empty);

    // Create a ray starting from the camera's position, going forward in the direction the camera is facing
    Ray ray = new Ray(cam.transform.position, cam.transform.forward);

    // Draw the ray visually in the Scene view for debugging (only visible in Editor, not in builds)
    Debug.DrawRay(ray.origin, ray.direction * distance);

    // Variable to store information about whatever the raycast hits
    RaycastHit hitInfo;

    // Fire the raycast. "out hitInfo" means the function will fill hitInfo with collision data if it hits something.
    // "distance" limits how far the ray checks, "mask" limits which layers it can detect (ignores irrelevant objects)
    if (Physics.Raycast(ray, out hitInfo, distance, mask))
    {
        // Check if the object we hit has an Interactable component attached
        if (hitInfo.collider.GetComponent<Interactable>() != null)
        {
            // Store a reference to that Interactable component
            Interactable interactable = hitInfo.collider.GetComponent<Interactable>();

            // Show that object's custom prompt message (e.g. "Press E to open door")
            playerUI.UpdateText(interactable.promptMessage);

            // Check if the Interact input action was pressed this frame
            if (inputManager.onFoot.Interact.triggered)
            {
                // Call the interact function on that object (opens door, picks up item, etc.)
                interactable.BaseInteract();
            }
        }
    }
}
```

Enemy detection using Raycast:

sightDistance = 20f

fieldOfView = 85f

```
public bool CanSeePlayer()
{
    if(player != null)
    {
        // is the player close enough to be seen?
        if (Vector3.Distance(transform.position, player.transform.position) < sightDistance)
        {
            Vector3 targetDirection = player.transform.position - transform.position - (Vector3.up * eyeHeight);
            float angleToPlayer = Vector3.Angle(targetDirection, transform.forward);
            // is the player within the field of view of enemy?
            if (angleToPlayer >= -fieldOfView && angleToPlayer <= fieldOfView)
            {
                Ray ray = new Ray(transform.position + (Vector3.up * eyeHeight), targetDirection);
                Debug.DrawRay(ray.origin, ray.direction * sightDistance);

                RaycastHit hitInfo = new RaycastHit();

                // is enemy's sight blocked by any object?
                if (Physics.Raycast(ray, out hitInfo, sightDistance))
                {
                    if (hitInfo.transform.gameObject == player)
                    {
                        return true;
                    }
                }

            }
        }
    }
    return false;
}
```